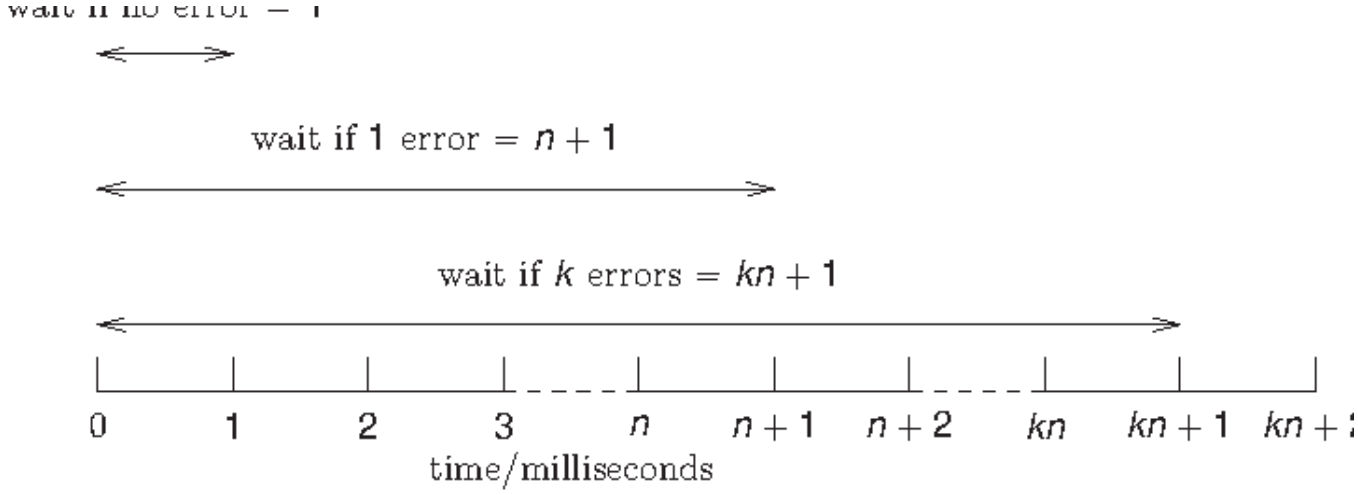
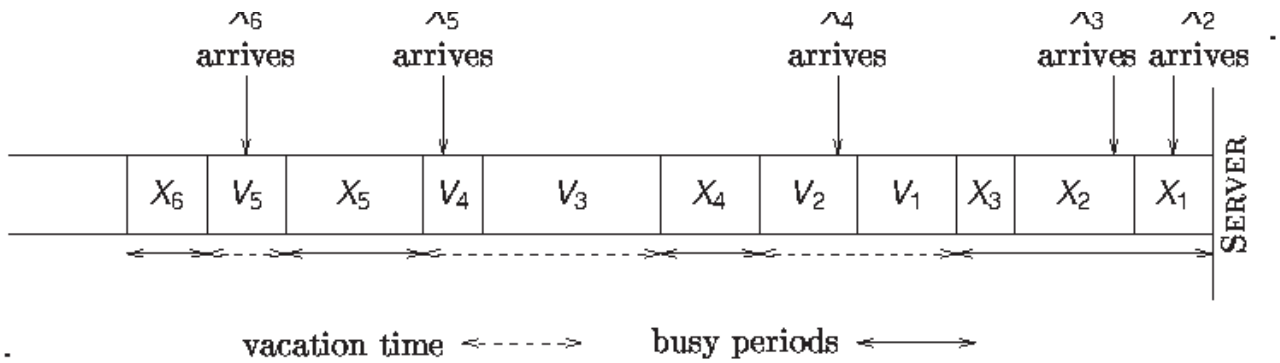
The M/G/1 Queue

2006-10-30

Graphical computation of mean residual time. $M(t)$ is the number of service completions in the interval $[0, t]$.



The ARQ System.



Schematic of $M/G/1$ queue

$M/G/1$ Queue

- Service times are *generally*distributed: cannot use the birth death process to analyse the system.
- We introduce a different type of analysis based on the mean residual service time.

$M/G/1$ Queue (continued)

- We will derive the Pollaczek-Khinchin formula.
- This gives the expected queueing time for any distribution of service times with *finite variance*.

Pollaczek-Khinchin formula

- The Pollaczek-Khinchin formula is given as

Pollaczek-Khinchin formula (continued)

- $$\bar{W} = \frac{\lambda \langle X^2 \rangle}{2(1-\rho)}$$

where * $\bar{W}$ is expected waiting time, * $\langle X^2 \rangle = \langle X^2 \rangle_{p(X)}$ is the second moment under the service time distribution, * utilisation factor is given as $\rho = \lambda \bar{X}$ and * $\bar{X} = \langle X \rangle_{p(X)}$ is the mean under the service time distribution.

Proof of Pollaczek-Khinchin Formula

- Proof of the Pollaczek-Khinchin formula.
- In our proof we will make three assumptions
- The queue is FIFO queue, *i.e.*customers are served in the order they arrive. This assumption may be relaxed but it will simplify our proof.

Proof of Pollaczek-Khinchin Formula (continued)

- The arrival times are statistically independent of the service times.
- The service time distribution has a variance which is finite.
- To formulate the proof we introduce a concept known as the mean residual service time.

The Residual Service Time

- The residual service time is . . . the amount of service time that remains for the customer under service from the perspective of a customer who has just arrived in the queue.
- The first part of the proof occurs from the perspective of the arriving customer.
- The arriving customer is considered to be the ith customer.

Simple Definitions and Schematic



residualServiceOverview

Simple Definitions and Schematic (continued)

- Example 1 in the Notes
- We introduce some simple definitions of terms.
- This enables us to draw a simple schematic of the system as it is perceived by the ith arrival. This is shown in Figure 12.
-

W_i	$=$	Waiting time in queue of ith customer.
R_i	$=$	'Residual service time' seen by the ith customer.
X_i	$=$	Service time of the ith customer.
k_i	$=$	Number of customers in the system when the ith customer arri

Wait for ith Customer

- 

residualServiceOverview

Wait for ith Customer (continued)

- From the figure the wait for the ith customer will be given by

$$W_i = R_i + \sum_{j=i-(k_i-1)}^{i-1} X_j$$

- Summing over both sides (*i.e.*for all the arriving customers up to the Ith)

$$\sum_{i=1}^{I} W_i = \sum_{i=1}^{I} R_i + \sum_{i=1}^{I} \sum_{j=i-(k_i-1)}^{i-1} X_j$$

- Dividing through by I gives us

$$\sum_{i=1}^{I} \frac{W_i}{I} = \sum_{i=1}^{I} \frac{R_i}{I} + \sum_{i=1}^{I} \sum_{j=i-(k_i-1)}^{i-1} \frac{X_j}{I}$$

- In the limit as $I \to \infty$ this becomes

Time Average Equivalence

- If we assume *time averages* are equivalent to *steady state averages* this gives

$$\langle W \rangle = \langle R \rangle + \langle (k-1)\, X \rangle \,.$$

- The length of the queue as seen by the ith arrival is independent of the ith arrival's length (our second assumption)

Time Average Equivalence (continued)

- this means $\langle (k-1)X \rangle = \langle k-1 \rangle \langle X \rangle$ and since $(k-1)$ is the length of the queue we may write

$$\bar{W} = \bar{R} + \bar{N}_Q \bar{X}$$

- Using Little's theorem for the queue we have

$$\bar{N}_Q = \lambda \bar{W}$$

- Substituting back in we have

$$\bar{W} = \bar{R} + \lambda \bar{X} \bar{W}.$$

Wait in Terms of Residual Service Time

- From the previous slide:

$$\bar{W} = \bar{R} + \lambda \bar{X} \bar{W}.$$

- And the fact that $\rho = \lambda \bar{X}$ (which stems from Little's formula for the service) gives us

$$\bar{W} = \bar{R} + \rho \bar{W}$$

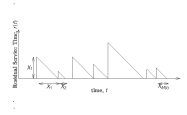
Wait in Terms of Residual Service Time (continued)

- Leading to

$$\bar{W} = \frac{\bar{R}}{1 - \rho}$$

- We now have the average queueing time in terms of the mean residual service time. To complete the Pollaczek-Khinchin formula we simply need to calculate $\bar{R}$.

Mean Residual Service Time from Departures Perspective



residualTime

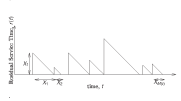
Mean Residual Service Time from Departures Perspective (continued)

- We again turn to a graphical proof to obtain the value of the mean residual service time.
-

Residual Service Time

- The time average of the mean residual service time can then be calculated in terms of the area under the curve, $A(t)$.
- This is given by the area under each of the triangles,

$$A(t) = \frac{1}{2} \sum_{i=1}^{M(t)} X_i^2.$$



residualTime

Residual Service Time

- The average service time is the area divided by time, t,

$$A(t) = \frac{1}{2}\sum_{i=1}^{M(t)} X_i^2.$$

$$\frac{A(t)}{t} = \frac{M(t)}{t}\frac{A(t)}{M(t)}$$

$$= \frac{1}{2}\frac{M(t)}{t}\frac{\sum_{i=1}^{M(t)} X_i^2}{M(t)}.$$

- we have introduced the number of arrivals after time t, as $M(t)$.
- As $t \to \infty$, *if the variance is finite*, we find that $M(t) \to \infty$ and we obtain

$$\bar{R} = \frac{1}{2}\lambda\left\langle X^2\right\rangle.$$

Pollaczek-Khinchin Formula

- Again assuming that time averages and ensemble averages are equal we substitute back into (17) to obtain the Pollaczek-Khinchin formula

$$\bar{W} = \frac{\lambda \langle X^2 \rangle}{2(1-\rho)}.$$

- This implies that the total waiting time in the system is

$$\bar{T} = \bar{X} + \frac{\lambda \langle X^2 \rangle}{2(1-\rho)}$$

and through Little's formula we can obtain the expected queue length, $\bar{N}_Q$, and the expected number in the system, $\bar{N}$.

Sanity Check — Markov Service times

- The exponentially distributed service times,

$$P(X) = \mu \exp(-\mu X),$$

have a mean of

$$\bar{X} = \frac{1}{\mu}$$

and a variance of $\frac{1}{\mu^2}$ leading to a second moment

$$\left\langle X^2 \right\rangle = \frac{2}{\mu^2}$$

- Substituting into the Pollaczek-Khinchin formula we obtain

$$\bar{W} = \frac{\rho}{\mu(1-\rho)}$$

Deterministic Service — $M/D/1$

- Some systems have deterministic service times. For example, transmission of fixed length packets over a channel with a fixed bandwidth.
- We represent a 'deterministic distribution' as

$$P(X) = \delta\left(X - \frac{1}{\mu}\right)$$

where $\delta(\cdot)$is the Dirac delta function. * For this system

$$\langle X \rangle = \frac{1}{\mu}$$

and

$$\left\langle X^2 \right\rangle = \frac{1}{\mu^2}$$

i.e. the variance is zero.

P-K Formula for Deterministic Systems

- For this system

$$\langle X \rangle = \frac{1}{\mu}$$

and

$$\left\langle X^2 \right\rangle = \frac{1}{\mu^2}$$

i.e. the variance is zero. * Substituting into the Pollaczek-Khinchin formula we obtain

$$\bar{W} = \frac{\rho}{2\mu\left(1-\rho\right)}.$$

Table of Delays and Average Numbers in System

- In fact we may derive the full range of results for these distributions.

Table of Delays and Average Numbers in System (continued)

Expected value	General	Exponential	Deterministic
$\bar{W}$	$\frac{\lambda}{2(1-\rho)}\langle X^2\rangle$	$\frac{\rho}{\mu(1-\rho)}$	$\frac{\rho}{2\mu(1-\rho)}$
$\bar{T}=\bar{W}+\frac{1}{\mu}$	$\frac{\lambda}{2(1-\rho)}\langle X^2\rangle+\frac{1}{\mu}$	$\frac{1}{\mu(1-\rho)}$	$\frac{2-\rho}{2\mu(1-\rho)}$
$\bar{N}_Q=\lambda\bar{W}$	$\frac{\lambda^2}{2(1-\rho)}\langle X^2\rangle$	$\frac{\rho^2}{(1-\rho)}$	$\frac{\rho^2}{2(1-\rho)}$
$\bar{N}=\lambda\bar{W}+\rho=$	$\frac{\lambda^2}{2(1-\rho)}\langle X^2\rangle+\rho$	$\frac{\rho}{1-\rho}$	$\frac{\rho(2-\rho)}{2(1-\rho)}$

Example — Uniform Service Times

- Uniform Service Times — $M/U/1$.
- Consider a uniform probability distribution is defined between 0 and b for $X > 0$ as

$$U(X|b) = \begin{cases} \frac{1}{b} & 0 < X \leq b \\ 0 & X > b \end{cases}$$

show that the man of this uniform distribution is given by $\frac{b}{2}$ and its variance by $\frac{b^2}{12}$ and find the expected waiting time for an $M/U/1$ queue?}

Answer

- For the mean we have

$$\langle X \rangle = \int_0^\infty X p(X)\, dX$$

Answer

- For the mean we have

$$\begin{aligned}\langle X \rangle = & \int_0^\infty X p\left(X\right) dX \\ = & \int_0^\infty X U\left(X|b\right) dX\end{aligned}$$

Answer

- For the mean we have

$$\begin{aligned}\langle X\rangle &= \int_0^\infty Xp\left(X\right)dX \\ &= \int_0^\infty XU\left(X|b\right)dX \\ &= \int_0^b \frac{X}{b}dX\end{aligned}$$

Answer

- For the mean we have

$$\begin{aligned}
\langle X \rangle &= \int_0^\infty X p\left(X\right) dX \\
&= \int_0^\infty X U\left(X|b\right) dX \\
&= \int_0^b \frac{X}{b} dX \\
&= \left[\frac{X^2}{2b}\right]_0^b
\end{aligned}$$

Answer

- For the mean we have

$$\begin{aligned}
\langle X \rangle &= \int_0^\infty X p(X)\, dX \\
&= \int_0^\infty X U(X|b)\, dX \\
&= \int_0^b \frac{X}{b} dX \\
&= \left[\frac{X^2}{2b}\right]_0^b \\
&= \frac{b^2}{2b} - 0
\end{aligned}$$

Answer

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$$
\begin{aligned}
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$$

Answer

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&= \frac{b^2}{2b} - 0 \\
&= \frac{b}{2}
\end{aligned}
$$

Variance

- For the variance we also require

$$\left\langle X^2 \right\rangle = \int_0^\infty X^2 p\left(X\right) dX$$

Variance

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$$\begin{aligned} \left\langle X^2 \right\rangle &= \int_0^\infty X^2 p\left(X\right) dX \\ &= \int_0^b \frac{X^2}{b} dX \end{aligned}$$

Variance

- For the variance we also require

$$\begin{aligned}\left\langle X^2 \right\rangle &= \int_0^\infty X^2 p\left(X\right) dX \\ &= \int_0^b \frac{X^2}{b} dX \\ &= \left[\frac{X^3}{3b}\right]_0^b\end{aligned}$$

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Variance

- For the variance we also require

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&= \int_0^b \frac{X^2}{b} dX \\
&= \left[\frac{X^3}{3b}\right]_0^b \\
&= \frac{b^3}{3b} - 0 \\
&= \frac{b^2}{3}
\end{aligned}$$

The variance is $\left\langle X^2 \right\rangle - \left\langle X \right\rangle^2 = \frac{b^2}{3} - \left(\frac{b}{2}\right) = \frac{b^2}{12}$.

Putting it Together

- For simplicity we consider $\mu = \frac{1}{\langle X \rangle}$. So $\mu = \frac{2}{b}$ and thus $b = \frac{2}{\mu}$.
- So we can write the second moment in terms of μ. $\langle X^2 \rangle = \frac{b^2}{3}$.

Putting it Together (continued)

- Substituting in $b = \frac{2}{\mu}$ we have $\langle X^2 \rangle = \frac{4}{3\mu^2}$.
- Substituting in to the Pollaczek-Khinchin formula we recover

$$\bar{W} = \frac{2\rho}{3\mu\left(1-\rho\right)}.$$

Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

Further Reading

- The $M/G/1$ queue, residual life, and vacations of Bertsekas and Gallager (1992)

Exercises

- Tutorial Sheet 3: triangular service time in Pollaczek–Khinchin of Lawrence (2006)

References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.

Lawrence, N.D., 2006. Erlang delay and loss and the M/G/1 queue.